Trace Tables

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Trace Tables before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Trace Tables trace and record the first step you can explain without help.

Use W2: **Trace** before running for **Loop**s and **trace table**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Trace Tables explanation against the supplied facts before discussing it.

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Trace Tables target to a trace and a justified conclusion.

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation. A student evaluates the condition using the initial value. Locate the divergence.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Trace Tables case and show the deciding evidence.

Set score=2; add 3; if score>4 subtract 1; double score.

Trace every value.

Show your **trace**, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

> Set score to 2. Add 3, then double the result. Record the score after each operation. What changes if you double before adding?

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 4 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	

Test record	My evidence
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Trace Tables adds to Transparent Book Recommender and keep the evidence.

Trace each book's genre, length, difficulty, and total score before sorting.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.

- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation. A student evaluates the condition using the initial value. Locate the divergence.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the range

> Explain the range. Apply the supplied Trace Tables facts and keep your reasoning.

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace before running

> Trace before running. Apply the supplied Trace Tables facts and keep your reasoning.

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair the reset

> Repair the reset. Apply the supplied Trace Tables facts and keep your reasoning.

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Build a repeated addition independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build a repeated addition independently. Apply the supplied Trace Tables facts and keep your reasoning.

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Check a contract limitation

> Check a contract limitation. Apply the supplied Trace Tables facts and keep your reasoning.

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Trace Tables

> A fictional reading-token jar starts with one token and gains three per round. Explain the labelled step before checking the changed condition.

A fictional reading-token jar starts with one token and gains three per round.

Python: total=1; for turn in range(4): total=total+3; after the **loop** print(total).

The update is inside the **loop**; print is outside.

AI ACADEMY / AI-2/4

Loop totals after each iteration

	turn	total after update
First	0	4
Second	1	7
Third	2	10
Fourth	3	13

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/4

Trace Tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial state

Before the loop, total=1.



2. Iteration values

range(4) supplies turn values 0,1,2,3.



3. Updated totals

After each turn total is 4,7,10,13.



4. After the loop

Print 13 once, after all four updates.

CHANGE ONE THING

Use range(0) instead of range(4).

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Move print inside the original **loop**. What sequence is displayed?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: **Trace** recommender **loop**s with current item, eligibility, score and current best so each update can be explained.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Trace recommender **loop**s with current item, eligibility, score and current best so each update can be explained.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional reading-token jar starts with one token and gains three per round.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Print 13 once, after all four updates.

> 1. **Iteration** values

2. Updated totals

3. After the **loop**

4. Initial **state**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: After the **loop** Print 13 once, after all four updates. The turn value labels an **iteration**; it is not the number added. The fixed update adds three each time.

Four **iteration**s take the total from 1 to 13. Keeping print outside the **loop** produces one final display rather than four intermediate displays. Initialise total before the **loop** so a zero-**iteration** run still has a defined value.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Use range(0) instead of range(4). Which conclusion is supported by the supplied facts?

1. The **loop** body never runs and the printed total is 1.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **loop** body never runs and the printed total is 1. The iterable contains no turn values; the initial assignment still happened.

Repair a missing building block

Review Branching Logic (ai-2/3). Goal: **Trace** the selected branch in an if elif else chain. Short practice: Give expected labels for –1,0,10,11 and identify the equality boundary for Short..

Return to this week and explain what you changed.

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If print(total) moves before total=total+2, what gets printed for the original **input**?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a **trace** rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

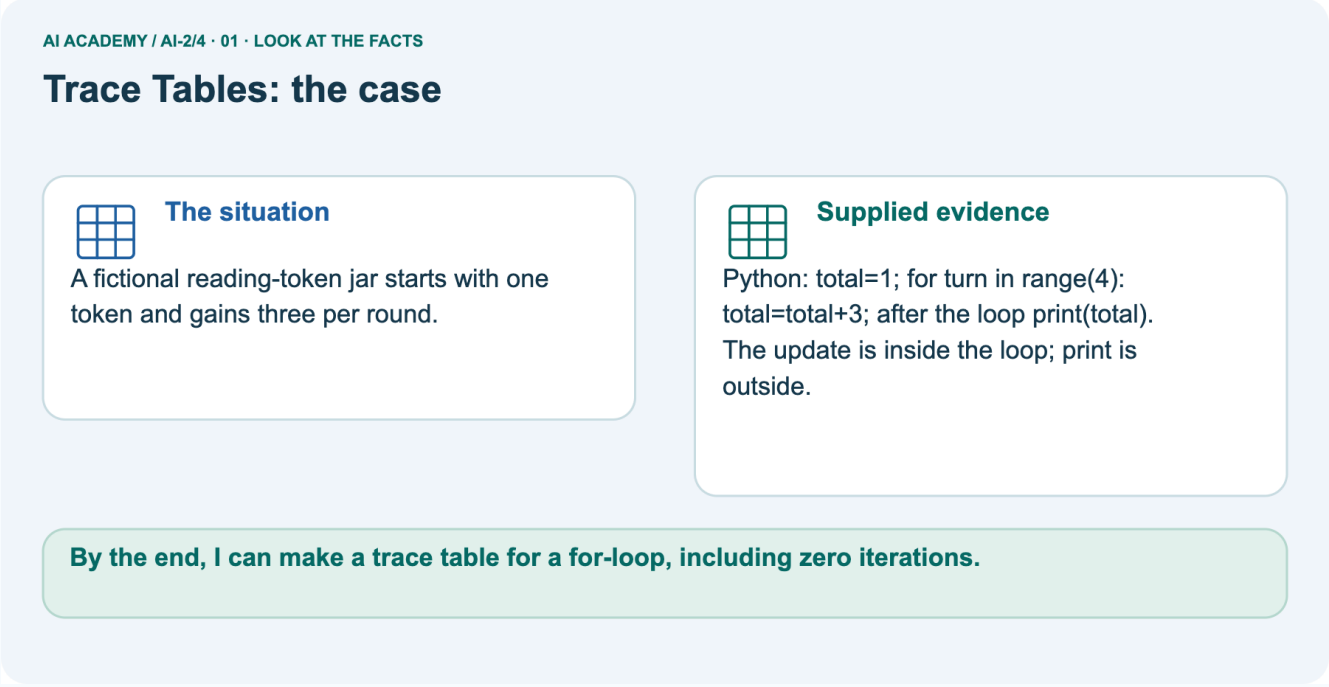
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Trace Tables picture from its labels, then record a relevant limit.

01. Trace Tables: the case



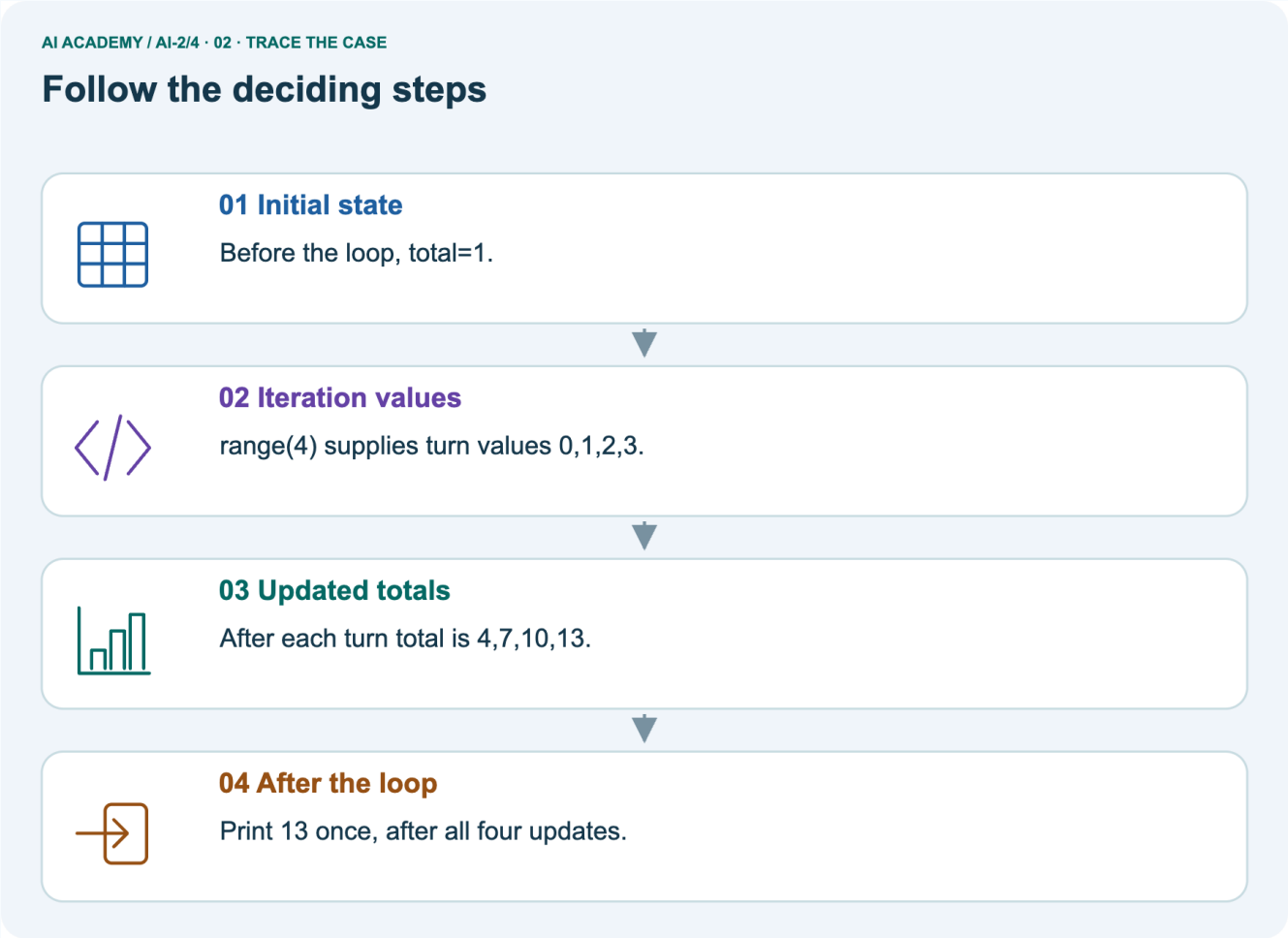
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional reading-token jar starts with one token and gains three per round. Supplied facts: Python: total=1; for turn in range(4): total=total+3; after the **loop** print(total). The update is inside the **loop**; print is outside.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Initial **state**: Before the **loop**, total=1. → **Iteration** values: range(4) supplies turn values 0,1,2,3. → Updated totals: After each turn total is 4,7,10,13. → After the **loop**: Print 13 once, after all four updates.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Loop totals after each iteration



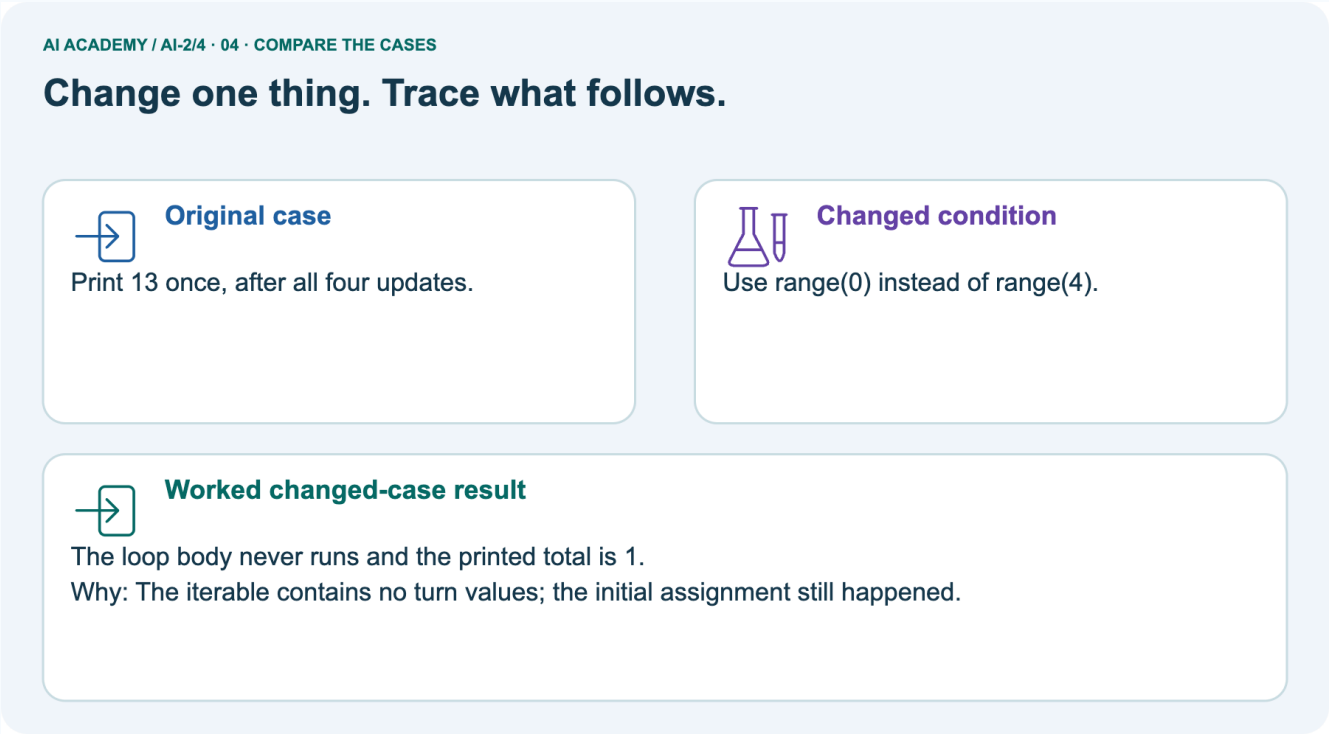
Follow the labelled picture with the matching explanation. Open the vector diagram

Loop totals after each **iteration**. First: turn = 0, total after update = 4. Second: turn = 1, total after update = 7. Third: turn = 2, total after update = 10. Fourth: turn = 3, total after update = 13.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

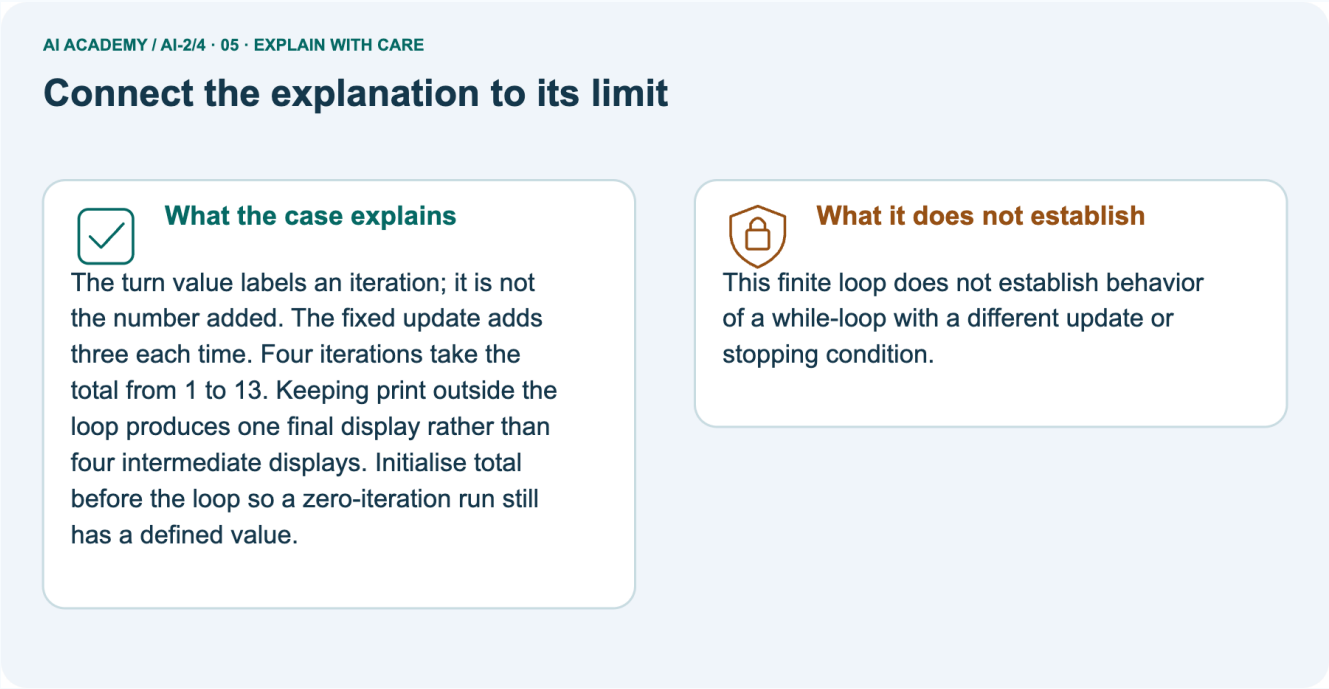
Original case: Print 13 once, after all four updates. Changed condition: Use range(0) instead of range(4). Expected result: The **loop** body never runs and the printed total is 1.

Reason: The iterable contains no turn values; the initial assignment still happened.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The turn value labels an **iteration**; it is not the number added. The fixed update adds three each time. Four **iteration**s take the total from 1 to 13.

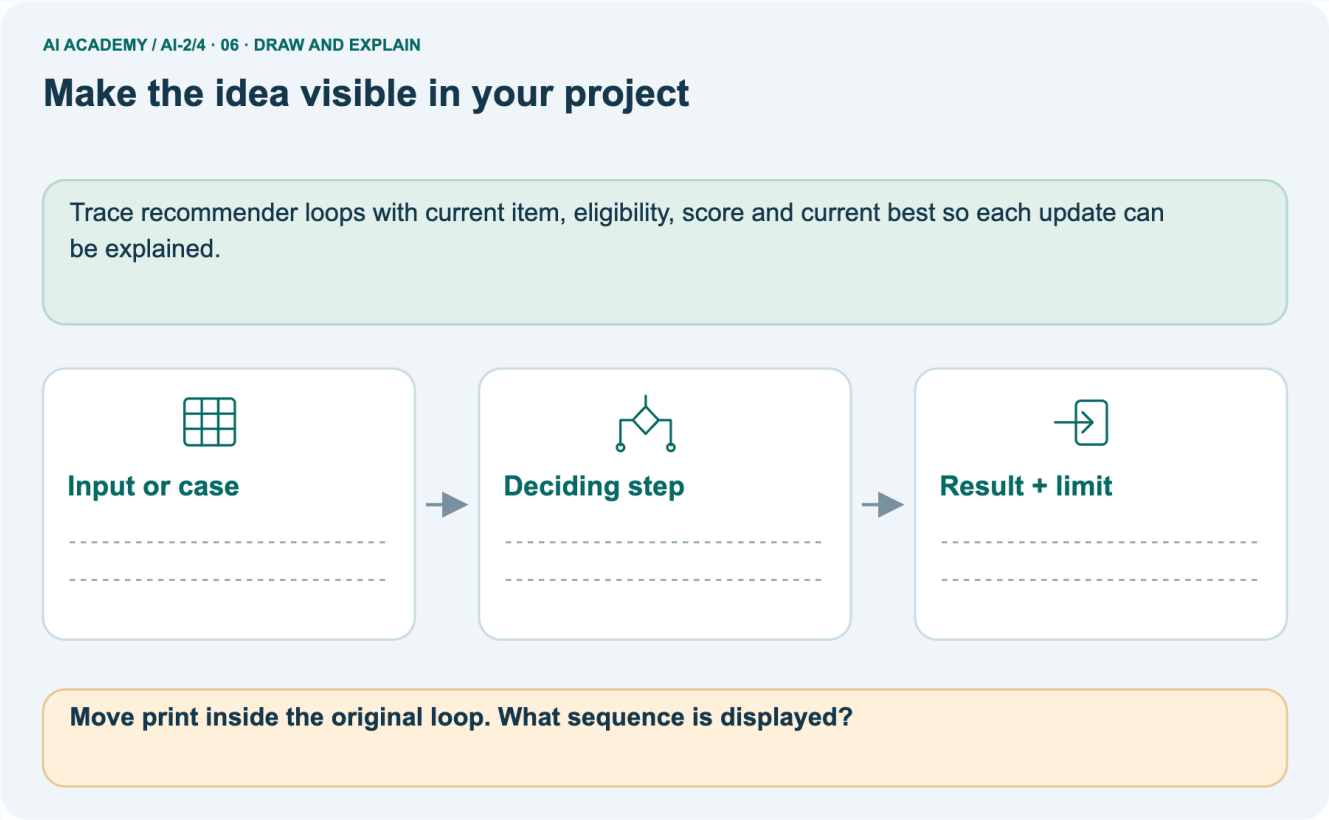
Keeping print outside the **loop** produces one final display rather than four intermediate displays. Initialise total before the **loop** so a zero-**iteration** run still has a defined value.

Limit: This finite **loop** does not establish behavior of a while-**loop** with a different update or stopping condition.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace recommender **loop**s with current item, eligibility, score and current best so each update can be explained. Draw your **input**, deciding step, result and limit. Supported question: Move print inside the original **loop**.

What sequence is displayed?

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.